

Procta

AI-Proctored Exam Platform

Feature inventory · as of 03 Jul 2026

What's actually shipped — sourced from the codebase, not the roadmap. Every entry below corresponds to working code in production today.

1. Live Proctoring Engine

Multi-signal AI runs in the locked Electron browser on the student's machine. Frames stream to the teacher dashboard in real time over WebSocket; events stream over Server-Sent Events.

Face presence	Detects when the student leaves the camera. InsightFace embedding model (CPU). Logged as <i>face_missing</i> violation.
Wrong-person detection	Embedding compared against the ID-verification selfie at calibration. Triggers if a different person sits at the keyboard mid-exam. <i>wrong_person</i> violation, weighted heaviest in risk score.
Multi-face detection	Two faces in frame for >2 s → <i>multiple_faces</i> violation. Catches another person looking over the shoulder.
Gaze tracking	MediaPipe face-mesh + 9-point calibration grid. Logs sustained off-screen gaze. Per-student calibration quality tier (Tight / Loose / Suspicious) shown in Results.
Head-pose estimation	Yaw / pitch / roll from face landmarks; gating threshold tuned in calibration to per-student baseline (hardware-aware).
Cheat-object detection (YOLOv8 + SAHI)	Phone, headphones, earbuds (custom ear-crop classifier). Slicing-aided inference for objects partially out of frame. GPU auto-detect: CUDA → MPS → CPU.
Audio anomaly (RMS)	Sustained voice / conversation-pattern detection via sounddevice RMS. Tuned to ignore typing and ambient noise. Three event types: <i>voice_detected</i> , <i>sustained_voice</i> , <i>conversation_detected</i> .
On-device speech-to-text keyword detection	Vosk (en-IN + hi-IN) runs on the student's CPU and matches transcripts against a built-in cheat-phrase list plus per-exam teacher-added keywords. Fires <i>keyword_uttered</i> on match with the matched phrase and a ~5 s transcript snippet. Audio never leaves the device.
Multi-voice detection	Silero VAD gates voice-active segments; python_speech_features MFCC vectors clustered over a rolling 60 s window via a numpy-only 2-cluster silhouette check. Two distinct voices in the buffer fires <i>multiple_voices_detected</i> . Catches an off-camera helper that RMS alone would miss.
Behavioural correlation engine	Multi-signal fusion: gaze + audio + face-pose patterns scored together to surface coordinated cheating (e.g. consistent gaze drift right + faint voice spikes).

2. Lockdown Browser (Electron)

The student's exam runs in a custom Electron build. The OS session is locked while the exam is active.

Tab-switch prevention	Browser focus loss for >1 s → <i>tab_switch</i> violation.
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Process integrity scan	Live scan for VPN, remote-desktop, screen-share, debugger, AI assistant tools. Hyphen-aware regex (e.g. parsec-cloud no longer false-positives parsec).
Hardware-locked session	Camera + microphone permissions granted at calibration; lost permission mid-exam ends the session.
Auto-update via electron-updater	Hits GitHub Releases on launch. Existing installs upgrade on next relaunch with no manual intervention.
Offline answer resilience	LocalStorage backup of every keystroke. On reconnect, background bulk-save catches up. Students never lose answers to a network blip.
Mac code-signed	Notarised .dmg distributed via /download/mac and /download/mac-x64. Windows code-signing pending Azure Trusted Signing.

3. Teacher Dashboard

Real-time control plane at app.procta.net. Single-page app with sticky chrome and single-scroll content per pattern.

Live sessions	47-student grid view with live face-frames over WebSocket, auto-updating risk scores, severity badges, force-submit, broadcast messages.
AI live-risk triage	One-line LLM summary per session, cached 60 s in Redis. Surfaces the 3 students most likely to be cheating right now.
Forensics timeline	Per-session scrub-through of every violation event with linked evidence screenshots. Downloadable PDF report.
Results table	Sortable, filterable, with per-row risk scores, calibration tier, and CSV / Excel / PDF / scorecard-ZIP exports.
Post-exam analytics	Score distribution (10 buckets), per-question difficulty + discrimination index, violation heatmap, risk band breakdown. Cached 60 s.
Real-time risk alerts	SSE toast notifications when a student's risk score crosses thresholds, with deep-link to the timeline.
Question editor + lint	MCQ / multi-select / true-false / short-answer types. AI lint runs each question pre-publish to surface ambiguity, leakage, or factual errors.
Question bank + import / export	Reusable pool of questions across exams. CSV / JSON bulk import with preview. Tag-based filtering.
Exam templates	Save an exam configuration + question set as a template; clone for next semester or section in one click.
Student groups / sections	Restrict exam access by group membership. Bulk add from registered students.
Live chat with students	Bidirectional WebSocket chat. Teacher broadcast or 1:1. Chat hub bounded with eviction (50 sockets/tenant, 4 h metadata TTL).
Live teacher intervention (Warn / Pause / Resume / End)	Three escalating actions on each live-session row. Warn pushes an amber banner + chime to the student with a chip-coded reason (eyes_off_screen / phone_visible / talking_to_someone / multiple_tabs / other). Pause locks the student UI with a full-screen overlay AND stops the exam timer; resume credits the paused interval back. End requires re-authentication and a chip-coded reason (academic_dishonesty / identity_fraud / environment_issue / repeated_violations / student_request / technical_failure / other); the reason persists on the session row, embeds in the audit-trail violation, and surfaces in the scorecard PDF + CSV export.

Cluster review (bulk false-positive triage)	Groups not-yet-dismissed violations by (exam, type, severity) across an entire cohort. One-click bulk dismiss with a reason stamps <code>dismissed_at</code> + <code>dismissed_reason</code> on every row in scope. Critical at 3,500-student exam scale where per-session review doesn't fit a teacher's afternoon.
Onboarding wizard	5-step intro for first-time teachers. Persists <code>procta_onboarded</code> in <code>localStorage</code> ; ? button re-opens.

4. Student Experience

Email invite → desktop install → calibration → exam. Calibration is mandatory and quality-scored.

Invite-driven enrolment	Teacher sends email invite; clicking the link auto-creates the student account on first <code>/validate-student</code> call. No pre-registration needed.
9-point gaze calibration	Mandatory before exam start. Quality tier baked into the session record so teachers can flag suspicious-tight calibrations later.
Continuous identity verification	Face embedding sampled throughout calibration; mismatch aborts the exam with a typed <code>calibration_abort</code> event that pages the teacher.
ID-document verification	Selfie + photo-ID upload. Stored as evidence; teacher reviews in dashboard pending-verifications panel.
In-exam chat with invigilator	Floating action button on the exam window. Encrypted, auth-scoped to the session.
Practice mode	Sandbox <code>session_id</code> (<code>PRACTICE_...</code>) that exercises the full flow against canned questions. No DB writes.
Multi-exam lobby	Single auth → list of available exams → pick one → calibrate → write.
Student performance history	Longitudinal view across all exams the student has taken with this teacher (when student account is connected).

5. Grading and Assessment

Auto-graded MCQ / multi-select / T-F	Scored at submit. Multi-select uses set-equality. Shuffled-option mappings preserved per-session for accurate grading.
Short-answer with AI-suggested grading	Teacher writes reference + rubric + max_score. LLM proposes score + 1-2 sentence rationale + confidence (high/med/low). Teacher confirms or overrides; only <i>teacher_score</i> counts in the gradebook.
Rolled-up scoring	On grade-confirm, MCQ correct + sum(teacher_score) auto-recomputes the session total + percentage. Idempotent — safe to re-run.
Per-student scorecard PDF	Header, score summary, per-question table (student answer vs reference, correct/incorrect), violation summary, calibration tier. When the session was force-submitted by a teacher, a Termination row surfaces the reason chip + the free-text note so the document is defensible in a later grade-challenge or DPDP review.
Bulk scorecard ZIP	Streams scorecards for an entire exam as a ZIP via BytesIO. Doesn't load all PDFs into memory.
Email scorecards	Idempotent per <i>scorecard_emailed_at</i> . Teacher can click twice; already-emailed students skipped.
Composite risk score (0–100)	Log-saturating per-violation-type weights, duration-normalised. Cached. Recomputable via admin backfill endpoint.
AI session narrative	1-paragraph human-readable summary of a student's session (LLM over the last 80 violation events). Cached 60 s.
Inbuilt coding questions	Server-side sandboxed execution (isolate, network-isolated, per-run CPU/memory/output caps) judges Python, JavaScript, TypeScript, C, C++, and Java against hidden tests. CodeMirror editor in the kiosk client; Run grades against sample cases only, Submit is judged server-side against hidden cases the student never sees inputs or expected outputs for.

6. Operations and Infrastructure

FastAPI + async stack	All hot-path endpoints async via <code>_atable</code> , backed by a native <code>asyncpg</code> connection pool (no Supabase dependency). Load-tested to 3,000 concurrent virtual students (p95 submit 51ms, 0.00% error rate) on a single Hostinger KVM VPS.
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Hardware governor (CPU-adaptive proctor cadence)	Reads CPU% via psutil every 5 s on the student machine. When CPU > 85% for two consecutive samples, ML inference drops to 1 frame per 2 s (event-only mode) and the audio worker skips every other ASR pass; rampback when CPU < 60%. Keeps the exam UI responsive on Rs. 30,000 budget Lenovo IdeaPad-class laptops instead of freezing under thermal throttling. Logs client_throttled (info severity) so teachers can correlate 'student exam felt slow' with hardware.
On-device model download bootstrap	scripts/download_audio_models.py invoked from Electron after pip install completes. Pulls Vosk en-IN + hi-IN + Silero VAD into ./weights/ with SHA-256 verification. Idempotent; cached after first run. Falls through to the existing RMS-only voice path if a download fails so the proctor never blocks on it.
Router decomposition	8 domain routers: auth / exam / admin / question_bank / grading / public / sse / chat. Single 7,000-line main.py shrunk to 382.
Pydantic strict-mode validation	ConfigDict(strict=True) on every request body model. Rejects type-coercion attacks (e.g. "123" for an int field) at the boundary.
Redis + in-memory caching	Two-layer: per-process LRU + Redis. Sorted-set LRU eviction for live frames (50-session cap). Cache-keyed risk scores, analytics, AI triage.
Atomic invite cap	Postgres RPC claim_invite_cap uses conditional UPDATE under row lock. Concurrent senders can't overshoot the daily cap.
Streaming CSV / PDF exports	StreamingResponse + async generators. Doesn't OOM on 5,000-row results tables.
Structured JSON logs + request IDs	Every request gets an X-Request-ID; logs are JSON for grep + log-aggregation tooling.
Sentry integration (optional)	SENTRY_DSN env var enables capture; off by default for local dev.
Security headers	CSP, HSTS, X-Frame-Options DENY, Permissions-Policy (camera + mic blocked at the app level for non-exam paths).
ETag + conditional requests	Middleware adds ETags to JSON responses ≤1 MB; 304 on matching If-None-Match.
GZip response compression	60-80% reduction on JSON >500 B. Static assets pre-gzipped at Docker build, served by Caddy.
Auto-update pipeline	GitHub Releases auto-discovered by backend (10 min TTL). Download buttons resolve to the latest release without redeploy.
Migration discipline	15 numbered SQL migrations, all idempotent. PGRST204 schema-cache drift handled with runtime fallback.

Test suite

14 pytest files: alerts, sessions, history, summary, behavioural analysis, e2e proctor, forensics timeline, invites, supporting modules. CI excludes heavy proctor tests via marker.

7. Communication and Workflow

Email invites with reminders	1 h + 24 h reminder cadence for unaccepted invites. Tracked via <code>opened_at</code> , <code>clicked_at</code> , <code>accepted_at</code> .
Real-time teacher↔student chat	Reconnection-tolerant WS. Per-session message history bounded by deque + cleanup loop.
Broadcast announcements	Teacher → all live students in an exam. Persists in chat history.
Force-submit + clear session	Time-windowed clear tokens (60 s TTL); audit-trailed violation event for every clear.
Webhook outbound (planned)	Schools' SIS systems. Stub exists; payload schema pending.

8. Privacy and Compliance Posture

RLS policies on all tables	Every table enforces <code>teacher_id</code> filtering at the DB level via native Postgres row-level security — the restricted <code>procta_app</code> runtime role can only see rows scoped by session-context functions (<code>app.teacher_id()</code> , <code>app.is_privileged()</code>). Even if the API has a bug, cross-tenant data is unreachable.
JWT-scoped session ownership	Session keys embed <code>teacher_id</code> UUID; access validated against JWT <code>tid</code> claim on every authenticated call.
Encrypted at rest and application-layer envelope encryption	Forensic evidence (screenshots) stored in AWS S3 (Mumbai, India data residency) with SSE-S3 encryption at rest. Exam answer keys and coding hidden test cases are additionally envelope-encrypted (AES-256-GCM) at the application layer before they ever reach the database, so a stolen database dump or a compromised DB role sees ciphertext, not answers.
HTTPS-only + HSTS	Caddy auto-provisions Let's Encrypt; HSTS preload-eligible.
Zero raw-video storage	Camera frames are processed on the student's device by the on-device ML stack. Only JPEG snapshots of flagged moments are uploaded as evidence; no continuous video reaches Procta servers. Materially reduces both the DPDP-aligned consent surface and the operating cost of storage.
Zero raw-audio storage	Microphone capture is analysed by the on-device Vosk + Silero VAD + MFCC pipeline. Only event metadata (<code>keyword_uttered</code> with the matched phrase, <code>multiple_voices_detected</code> with a confidence score) and the synchronously-captured camera JPEG are uploaded. Raw audio never leaves the student's machine.

Auto-delete evidence retention	Cron in entrypt.sh auto-purges evidence JPEGs older than SCREENSHOT_RETENTION_DAYS (default 90). Aligned with DPDP storage limitation and the common 90-day exam-record retention norm in Indian higher education.
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9. Roadmap Snapshot

Items with working code or migrations queued. Prioritised by school-procurement leverage.

Windows code signing	Certum EV code-signing certificate (~\$290 per ~15-month issuance, per the 2026 CA/Browser Forum validity cap). Removes the SmartScreen warning immediately on install — currently a deal-blocker for school IT evaluating an unsigned installer.
Mobile PWA (lite proctoring)	Camera-based proctoring still works; tab-switch prevention doesn't. Position as low-stakes-quiz tool.
Per-criterion rubric grading	Extend short-answer to (clarity / correctness / depth) × marks. Universities specifically asked.
Cross-cohort plagiarism check	Cosine sim on embeddings of every short-answer pair. Cheap with a small embedding model.
Two-factor for teacher login	TOTP via authenticator app.
Bulk exam scheduling	Clone exam to N sections × M time slots. Coaching-institute ergonomic win.

10. Future Vision

Bigger bets — 6–24 months out. Not on the immediate sprint, but the direction Procta evolves to become a category-defining platform rather than a proctoring tool.

React dashboard rewrite	The current 7,000-line vanilla-JS dashboard is functional but ages every time a feature lands. Migrate to a React + Vite SPA matching the marketing site's design system. Wins: component reuse, dark/light theme switching, keyboard shortcuts, fluid loading states, drag-to-reorder, auto-escaping (kills the entire XSS class), and a real design language for sales screenshots.
Self-hostable on-prem coding-execution appliance	Coding-question execution ships today as a Procta-hosted service (see Section 5). A packaged, self-hostable appliance variant would let a university's own IT run the sandbox on-prem for institutions with a no-external-code-execution policy. Stronger VM-level isolation (Firecracker microVM, in addition to the current isolate sandbox) was explored for this but is blocked on hosting: it needs a host that exposes nested KVM virtualization, which the current production VPS does not.
AI exam generator from syllabus	Upload a PDF syllabus or paste a topic list; LLM produces a balanced 40-question exam tagged by Bloom's level + difficulty. Teacher tweaks → publish. Saves the 2-hour exam authoring session that's currently every teacher's worst job.
On-device speaker identification (deferred)	Today's multi-voice detection answers "two voices yes/no" via MFCC clustering. Speaker identification would add a stable label per voice ("voice A" / "voice B") and track when each appears. Requires either a pyannote ONNX export (~30 MB additional model) or a custom siamese-network fingerprint head. Worth doing only if customer feedback shows the yes/no signal is leaving evidence on the table.
Native mobile apps (iOS + Android)	React Native shell wrapping the same proctoring engine compiled for ARM. Position as the primary tool for low/medium-stakes assessments where the desktop install is friction. Camera-based proctoring still works; lockdown is weaker but compensated with stricter face-presence policy.
Teacher mobile app	Push-notify on high-risk events. Swipe-right to acknowledge, swipe-left to force-submit. For invigilators monitoring remotely. Pairs with the desktop dashboard, doesn't replace.
LMS integrations	Canvas, Moodle, Blackboard, Google Classroom plugins. Single-sign-on, exam launch from inside the LMS, scores auto-pushed back. The integration story is what unblocks university procurement deals.

Adaptive testing (IRT-based)	Questions adjust difficulty to the student's demonstrated level mid-exam, using item-response theory. Same total time, more accurate measurement, harder to brute-force-cheat (no two students see the same question sequence).
Teacher AI assistant	Chat with your gradebook in natural language: <i>"Which students improved most this semester?" "Show me question #14's discrimination index across all sections."</i> Backed by an LLM with read-only tool access to the teacher's data.
AI proctor explainer for students	When a student is flagged, they see a plain-English explanation: <i>"Your gaze drifted off-screen for 12 s during Q14, which is longer than 95% of test-takers."</i> Removes the "proctoring is a black box" hostility.
Cross-cohort plagiarism check	Cosine similarity on embeddings of every short-answer pair across the cohort. Highlights suspicious clusters. Cheap with a small embedding model run nightly.
Live invigilator marketplace	When AI flags a high-risk session, optionally route to a human reviewer (vetted, on demand, paid per minute). Schools that don't have invigilator capacity buy review time from the platform. Two-sided revenue stream.
Multi-monitor detection	Browser screen-API + webcam reflection analysis to detect secondary monitors. Currently the loophole every prep-school student exploits.
rPPG attention + stress metrics	Heart-rate variability inferred from sub-pixel facial colour changes (research-grade rPPG). Surface attention dropoff and stress to teachers without storing biometric data — computed live, never persisted.
Whiteboard / stylus mode	Students show working on iPad / Wacom tablet for math + physics exams. Stroke-by-stroke recording for grading and review. OCR + LLM for partial credit on handwritten derivations.
Federated SSO for school districts	SAML / OIDC integration so a district admin manages 50 schools' teachers from one console. Critical for K-12 district sales.
Public API + developer SDK	Third-party integrations on top of Procta — e.g. coaching institutes plugging Procta into their custom LMS. Rate-limited, scoped tokens, OpenAPI spec. Becomes a moat.
On-device LLM for free tier	Llama / Phi running locally for grading, lint, and risk narratives. Removes Groq dependency and cuts inference cost to zero for paying schools. Slower but private — useful selling point in EU.
Exam marketplace	Teachers publish question banks; others subscribe. Teachers earn revenue on usage. 30/70 split, Procta keeps 30%. Compounds the network effect once question-bank UX matures.

Generated from the Procta codebase. Sections 1–8 list features with shipping code or tracked migrations. Section 9 is near-term roadmap (working code or designs queued). Section 10 is directional — bigger bets that define where Procta goes next.